

NI TB Genomic Surveillance

Outbreak Investigation Report

Generated: 2026-05-08 14:13:08 UTC

Case Summary

Total Cases	250
Clustered Cases	171
Unclassified Cases	79
Open Clusters	0

Automated Interpretation Flags

- Sequencing coverage is below target: 0.0% (target >= 80%).
- High-confidence transmission links present; prioritize epidemiology follow-up.

Analysis Summary

Posterior Samples	2000
MCMC Iterations	2500
Burn-in	500
Mean Log-Likelihood	-5250.50
Log-Likelihood SD	145.30
Transmission Probability	0.085
Generation Time Mean (days)	12.50
Generation Time SD (days)	3.200
Sampling Probability	0.650
Convergence Diagnostic	Rhat < 1.1 (good convergence)

Programme Surveillance KPIs (Last 12 Weeks)

Eligible Cases	30
Sequenced Cases	0
Sequencing Coverage (%)	0.0
QC Reported Cases	0
QC Pass Cases	0
QC Fail Cases	0
QC Pass Rate (%)	None
Contamination Flags	0
Median Days Specimen to QC	None

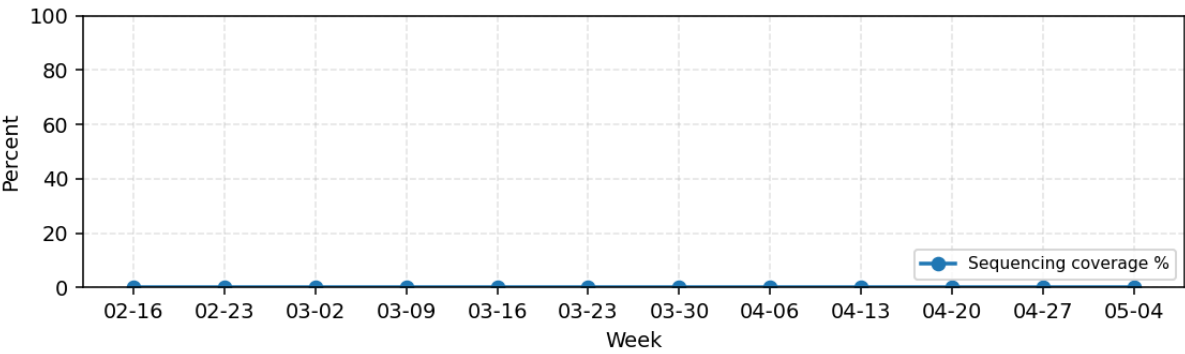
KPI Warning: sample_qc_metrics table not found; apply updated db/schema.sql to enable QC KPIs.

Regional Sequencing Representativeness

Region	Eligible	Sequenced	Coverage %
United Kingdom	30	0	0.0

Weekly Surveillance Trends (12 Weeks)

Week	Eligible	Sequenced	Coverage %	QC Pass %
2026-02-16	3	0	0.00	None
2026-02-23	5	0	0.00	None
2026-03-02	1	0	0.00	None
2026-03-09	3	0	0.00	None
2026-03-16	4	0	0.00	None
2026-03-23	1	0	0.00	None
2026-03-30	4	0	0.00	None
2026-04-06	4	0	0.00	None
2026-04-13	2	0	0.00	None
2026-04-20	0	0	None	None
2026-04-27	1	0	0.00	None
2026-05-04	1	0	0.00	None



Cluster Action Prioritization

Cluster	Cases	Regions	Most Recent	Status	Priority
218dc8c3	39	1	2026-04-28	monitoring	84
0bc4e972	28	1	2026-05-06	monitoring	62
45a9c843	27	1	2026-04-09	closed	59
a1c1bbde	27	1	2026-03-29	monitoring	58
656390cf	27	1	2026-03-12	monitoring	58
e4277d16	23	1	2026-04-12	monitoring	51

Priority score combines cluster size, cross-region spread, specimen recency, and open investigation status.

Transmission Priority Signals

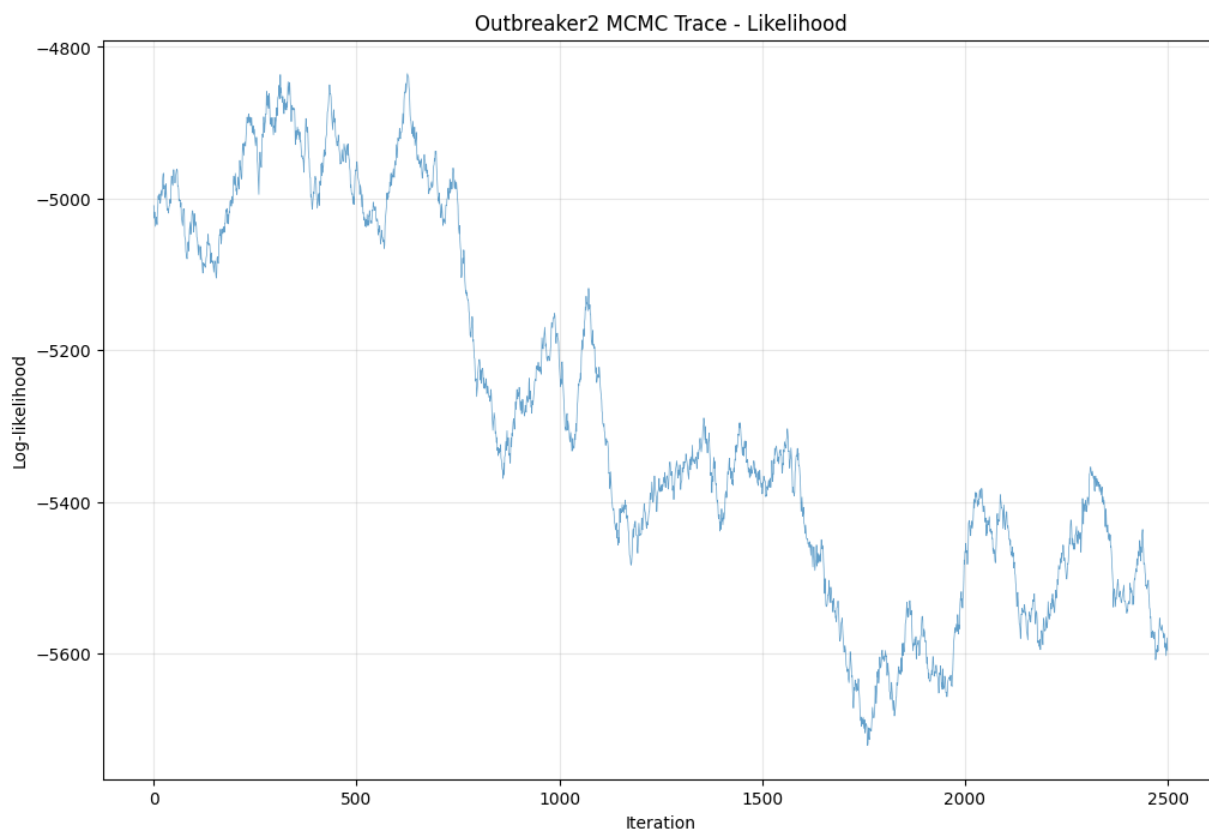
Case	Region	Risk	Out	In
e44cfc7c	United Kingdom	1.628	2	2

655aacc2	United Kingdom	1.622	2	2
0c0e07a0	United Kingdom	1.616	2	2
b3bc35ab	United Kingdom	1.606	2	2
31f9e48a	United Kingdom	1.603	2	2
625e44a8	United Kingdom	1.596	2	2
b0d54caa	United Kingdom	1.596	2	2
000bb4b4	United Kingdom	1.594	2	2
c27a2afa	United Kingdom	1.591	2	2
85a0d48b	United Kingdom	1.59	2	2

Network snapshot: nodes=171, edges=324, high-confidence links=136

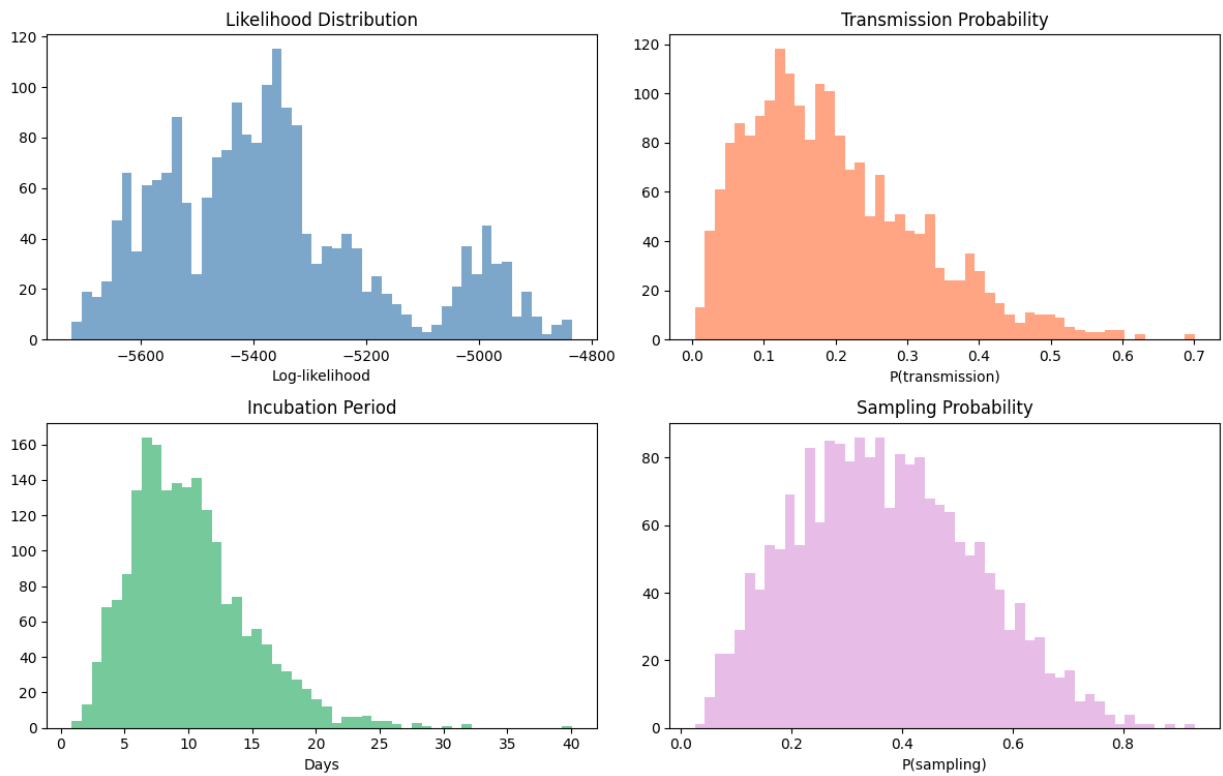
Diagnostic Graphics

Trace



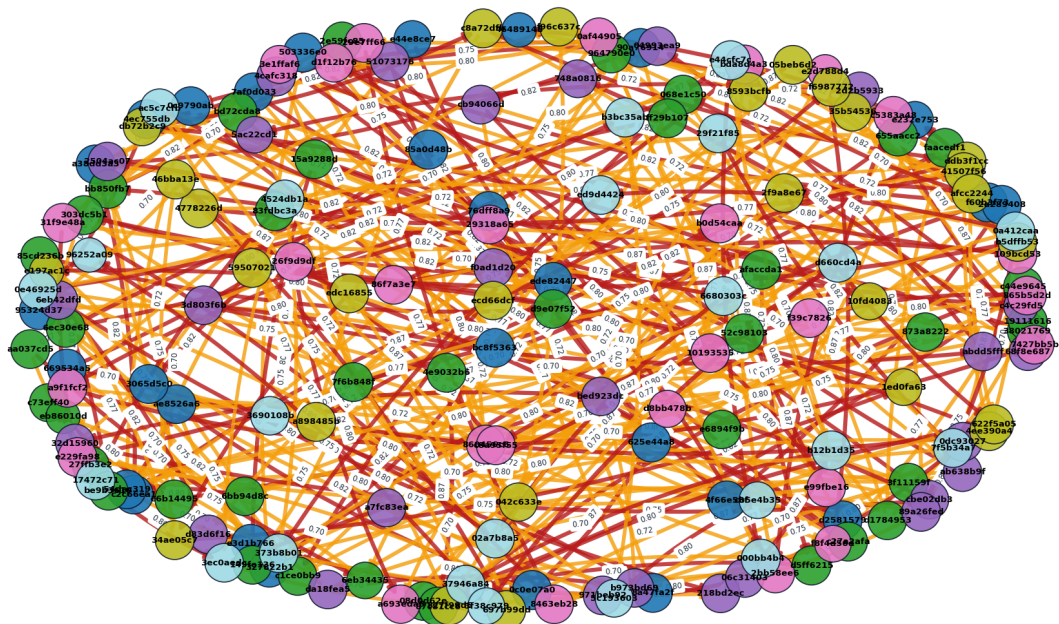
Hist

Outbreaker2 Posterior Distributions

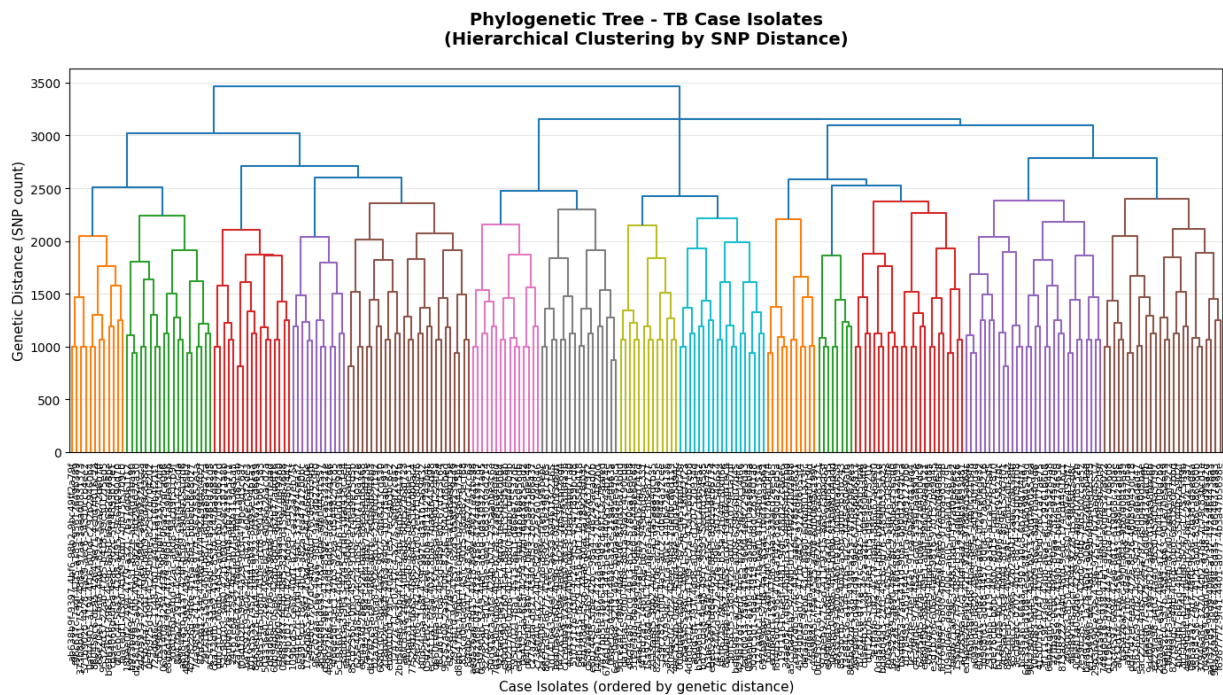


Tree

Enhanced Transmission Network
Node size = inferred spread risk | Edge label = transmission probability



Phylo



Resistance



Data Provenance

This report combines case tables, cluster assignments, outbreaker summary metrics, surveillance KPIs, and transmission network outputs available at generation time.

Outbreaker summary timestamp: 2026-05-08T15:09:15.252116

Transmission network timestamp: 2026-05-08T14:09:20